

```

11. other, produce the number. For example,  $2 \times 13 = 26$ , so 2 and 13 are
12. factors of 26.  $1 \times 26 = 26$ , so 1 and 26 are also factors of 26. We
13. say that 26 has four factors: 1, 2, 13, and 26.
14.
15. If a number only has two factors (1 and itself), we call that a prime
16. number. Otherwise, we call it a composite number.
17.
18. Can you discover some prime numbers?
19. '''
20.
21. while True: # Main program loop.
22.     print('Enter a positive whole number to factor (or QUIT):')
23.     response = input('> ')
24.     if response.upper() == 'QUIT':
25.         sys.exit()
26.
27.     if not (response.isdecimal() and int(response) > 0):
28.         continue
29.     number = int(response)
30.
31.     factors = []
32.
33.     # Find the factors of number:
34.     for i in range(1, int(math.sqrt(number)) + 1):
35.         if number % i == 0: # If there's no remainder, it is a factor.
36.             factors.append(i)
37.             factors.append(number // i)
38.
39.     # Convert to a set to get rid of duplicate factors:
40.     factors = list(set(factors))
41.     factors.sort()
42.
43.     # Display the results:
44.     for i, factor in enumerate(factors):
45.         factors[i] = str(factor)
46.     print(', '.join(factors))

```

Exploring the Program

Try to find the answers to the following questions. Experiment with some modifications to the code and rerun the program to see what effect the changes have.

1. What happens if you delete or comment out `factors.append(i)` on line 36?
2. What happens if you delete or comment out `factors = list(set(factors))` on line 40? (Hint: Enter a square number such as 25 or 36 or 49.)
3. What happens if you delete or comment out `factors.sort()` on line 41?